

Hershdeep Singh Chopra (Hersh Chopra)

Chicago, Illinois, USA

(424) 832-0952, hershdeepchopra2026@u.northwestern.edu

hersh42.github.io

EDUCATION

Northwestern University

PhD in Economics

Expected June 2026

University of California, Los Angeles (UCLA)

Bachelor of Science in Mathematics and Economics (with honors)

[GPA: 3.8]

June 2020

RESEARCH

Employer Competition and Certification (Job Market Paper)

Abstract: This paper develops a theory of employer competition over hiring standards in labor markets where employers rely on third-party certification to screen applicants. The certifier sells tests to an applicant, who possesses noisy private information about his ability and seeks to persuade employers to offer him employment. An employer hires if, conditional on the test outcome, her posterior belief about the applicant's ability meets her standard. Employers compete in their *selectivity* or *hiring standards* to attract the applicant. The certifier faces both a screening problem, due to applicants' private information, and an information design problem, as tests must be informative enough to persuade employers. The screening frictions faced by the certifier affect the equilibrium determination of hiring standards and the certification mechanism. I show that this inefficiency in test allocation leads to two distortions relative to a benchmark where the certifier only faces an information design problem. First, there is an increase in the probability of unemployment. The certifier's optimal selling mechanism leads to the *exclusion* of some applicant types. Second, competition among employers intensifies. The inefficient test allocation steers the applicant supply towards less selective employers, resulting in equilibrium *constriction of standards*.

Incentive-Compatible Information Design (joint with Jeffrey Ely)

Abstract: We study the design of mechanisms by an intermediary that generates information for a sender to persuade a receiver about an unknown attribute of the sender. The sender is initially privately, but imperfectly, informed about her attribute, and the receiver takes an action based on posterior beliefs about the sender's attribute and the

sender's belief about the attribute. The design of the mechanism, therefore, confronts both incentive-compatibility constraints (for the sender) and obedience constraints (for the receiver). We characterize profit-maximizing mechanisms when the intermediary contracts with the sender, and we specialize to two applications: the design of college-admissions tests and the optimal use of consumer data on a digital market platform. We also show how our analysis extends beyond a profit-maximizing intermediary by analyzing the design of budget-balanced selling mechanisms in a lemons market.

Interim Information Design (joint with Jeffrey Ely)

Abstract: We study efficient and consumer-surplus maximizing information policies in a bilateral trade setting where the buyer is initially privately imperfectly informed about his willingness to pay. We show that providing more information to the buyer can lead to higher market prices without affecting the consumer or producer surplus.

ACADEMIC EXPERIENCE

Research assistant (Interim and Incentive-compatible Information Design) for Professor Jeffrey Ely at Northwestern University, 2024-2025.

Research assistant (Lecture Notes on Information Economics) for Professor Annie Liang at Northwestern University, Summer and Fall 2022.

Research assistant (Learning Dynamics in Social Networks) for Professor Simon Board and Professor Moritz Meyer-ter-Vehn at UCLA, 2018 - 2019.

TEACHING EXPERIENCE

Teaching Assistant for Economics of Risk and Uncertainty (Econ 331) at Northwestern University under Professor Marciano Siniscalchi in Fall 2024.

Teaching Assistant for Game Theory (Econ 380) at Northwestern University under Professor Asher Wolinsky in Fall 2023.

Teaching Assistant for Economics of Data (Econ 313) at Northwestern University under Professor Annie Liang in Spring 2023.

Teaching Assistant for Intermediate Microeconomics II (Econ 310-2) at Northwestern University in Winter 2024, Spring 2022, and Fall 2021 under various faculty members.